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TREND-BASED MOCK PAPER

GATE Ecology and Evolution

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Ecology and Evolution
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Antonym of 'indigenous':

(Numerical Answer Type — Enter the numerical value)

Section B: Ecology and Evolution

Q2. [1-MCQ] [1 mark]

Standard deviation is:

- (A) Mean
 - (B) Spread
 - (C) Median
 - (D) Mode
-

Q3. [1-NAT] [1 mark]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
 - (B) 0.5
 - (C) 0.6
 - (D) 0.4
-

Q5. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only

- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q6. [1-MCQ] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q7. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q8. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q10. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q11. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q12. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q13. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q14. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q15. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q16. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q17. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q18. [2-MSQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q19. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q20. [2-NAT] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q21. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
 - (B) e^2
 - (C) 1
 - (D) infinity
-

Q22. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q23. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
 - (B) $2\pi r$
 - (C) πr
 - (D) $2\pi r^2$
-

Q24. [2-MCQ] [2 marks]

$d/dx [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
 - (B) 2
 - (C) 4
 - (D) 1
-

Q25. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = Cx$
 - (B) $y = C/x$
 - (C) $y = Cx^2$
 - (D) $x^2 + y^2 = C$
-

Q26. [2-NAT] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MSQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q28. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q29. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q30. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q31. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q32. [1-MCQ] [1 mark]

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

Q33. [1-MCQ] [1 mark]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q34. [1-MCQ] [1 mark]

ABO blood group has how many alleles?

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Q35. [1-MCQ] [1 mark]

Linked genes show:

- (A) Independent assortment
- (B) No recombination
- (C) Less than 50% recombination

(D) Always 50%

Q36. [1-NAT] [1 mark]

DNA B-form base pairs per turn:

(Numerical Answer Type — Enter the numerical value)

Q37. [1-MCQ] [1 mark]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q38. [2-MCQ] [2 marks]

ABO blood group has how many alleles?

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Q39. [2-NAT] [2 marks]

DNA B-form base pairs per turn:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MSQ] [2 marks]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q41. [2-MCQ] [2 marks]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q42. [2-MCQ] [2 marks]

Linked genes show:

- (A) Independent assortment
- (B) No recombination
- (C) Less than 50% recombination
- (D) Always 50%

Q43. [2-MSQ] [2 marks]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q44. [1-MCQ] [1 mark]

Photosynthesis light reactions produce:

- (A) Glucose only
- (B) ATP and NADPH
- (C) CO₂
- (D) O₂ only

Q45. [1-MCQ] [1 mark]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
- (B) 9:3:3:1
- (C) 1:2:1
- (D) 1:1:1:1

Q46. [1-MCQ] [1 mark]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q47. [1-MCQ] [1 mark]

Competitive enzyme inhibition: K_m

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Zero

Q48. [1-NAT] [1 mark]

DNA replication in eukaryotes occurs in:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-MCQ] [2 marks]

Competitive enzyme inhibition: K_m

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Zero

Q50. [2-NAT] [2 marks]

DNA replication in eukaryotes occurs in:

(Numerical Answer Type — Enter the numerical value)

Q51. [2-NAT] [2 marks]

Photosynthesis light reactions produce:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MCQ] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1

- (B) 9:3:3:1
- (C) 1:2:1
- (D) 1:1:1:1

Q53. [2-MCQ] [2 marks]

Competitive enzyme inhibition: K_m

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Zero

Q54. [1-MCQ] [1 mark]

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

Q55. [1-MCQ] [1 mark]

Most abundant greenhouse gas:

- (A) CO₂
- (B) CH₄
- (C) N₂O
- (D) Water vapor

Q56. [1-NAT] [1 mark]

Trophic level energy transfer efficiency:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

Logistic growth equation has carrying capacity:

- (A) $dN/dt = rN$
- (B) $dN/dt = rN(1-N/K)$
- (C) $dN/dt = r$
- (D) $dN/dt = K-N$

Q58. [2-MCQ] [2 marks]

Most abundant greenhouse gas:

- (A) CO₂
- (B) CH₄
- (C) N₂O
- (D) Water vapor

Q59. [2-NAT] [2 marks]

Logistic growth equation has carrying capacity:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

Darwin finches from:

- (A) Galapagos
- (B) Africa

- (C) Asia
- (D) Australia

Q61. [1-NAT] [1 mark]

Biodiversity hotspot criteria:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

Q63. [1-MCQ] [1 mark]

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

Q64. [1-MCQ] [1 mark]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q65. [1-MCQ] [1 mark]

Monarch butterfly migrates from:

- (A) Canada to Mexico
 - (B) India
 - (C) Africa
 - (D) Europe
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	D	1
6	1NAT	C	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1MCQ	A	1
10	1MCQ	A	1
11	1MCQ	D	1
12	1MCQ	B	1
13	1NAT	A	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1MCQ	A	1
17	1MCQ	B	1
18	1NAT	B	1
19	1MCQ	B	1
20	1NAT	B	1
21	1MCQ	B	1
22	1MCQ	C	1
23	1MCQ	B	1
24	1NAT	B	1
25	1MCQ	A	1
26	1MCQ	C	1
27	1MCQ	B	1
28	1MCQ	A	1

29	1NAT	A	1
30	1MCQ	A	1
31	2MCQ	A	2
32	2MCQ	A	2
33	2MCQ	A	2
34	2MCQ	B	2
35	2MCQ	C	2
36	2MCQ	C	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MSQ	C	2
40	2MCQ	A	2
41	2NAT	A	2
42	2MCQ	B	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	C	2
46	2MCQ	B	2
47	2NAT	A	2
48	2MSQ	A	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2NAT	B	2
52	2MCQ	A	2
53	2MSQ	A	2
54	2MCQ	A	2
55	2NAT	B	2
56	2MCQ	A	2
57	2MCQ	C	2
58	2MCQ	C	2

59	2MSQ	A	2
60	2NAT	B	2
61	2MCQ	D	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2NAT	B	2
65	2NAT	C	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Statistics | Difficulty: difficult

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Ecology | Topic: Population | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Evolution | Topic: Natural Selection | Difficulty: easy

Q5.

Answer: D

Solution:

Correct Answer: D

Subject: Genetics | Topic: Population Genetics | Difficulty: moderate

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Environmental Science | Topic: Biodiversity | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Cell Biology | Topic: Mitochondria | Difficulty: difficult

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Plant Physiology | Topic: Photosynthesis | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Zoology | Topic: Taxonomy | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Behavioral Ecology | Topic: Migration | Difficulty: easy

Q11.

Answer: D

Solution:

Water vapor: most abundant GHG by concentration and radiative forcing.

Subject: Ecology | Topic: Ecology | Difficulty: difficult

Q12.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q13.

Answer: A

Solution:

10% law: ~10% energy transferred between trophic levels (Lindeman).

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q14.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q15.

Answer: A

Solution:

Reverse transcription: RNA->DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q16.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q17.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q18.

Answer: B

Solution:

|x| is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: B

Solution:

ABO: I^A, I^B, i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q20.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q21.

Answer: B

Solution:

Logistic: $\frac{dN}{dt} = rN(1-N/K)$. K is carrying capacity.

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q22.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q23.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q24.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q25.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q26.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}*[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q44.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q47.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

$L\{t*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: B

Solution:

ABO: I^A , I^B , i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q51.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q52.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q53.

Answer: A

Solution:

$AaBb \times aabb$: offspring $AaBb$, $Aabb$, $aaBb$, $aabb$ in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q54.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q55.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q56.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q58.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q59.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q60.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: difficult

Q61.

Answer: D

Solution:

Water vapor: most abundant GHG by concentration and radiative forcing.

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q62.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: easy

Q63.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q64.

Answer: B

Solution:

Logistic: $\frac{dN}{dt} = rN(1-N/K)$. K is carrying capacity.

Subject: Ecology | Topic: Ecology | Difficulty: difficult

Q65.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 30 questions, 52 marks
Genetics: 12 questions, 18 marks
Biology: 10 questions, 15 marks
Ecology: 6 questions, 8 marks
General Aptitude: 1 questions, 1 marks
Evolution: 1 questions, 1 marks
Environmental Science: 1 questions, 1 marks
Cell Biology: 1 questions, 1 marks
Plant Physiology: 1 questions, 1 marks
Zoology: 1 questions, 1 marks
Behavioral Ecology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 18 | Moderate: 35 | Difficult: 12

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.